Indian Institute of Technology Guwahati
Department of Physics
PH102
Tutorial-4

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1. Suppose the electric field in some region is found to be $\vec{E} = kr^3\hat{r}$, in spherical coordinates (k is some constant).
   (a) Find the charge density $\rho(\vec{r})$.
   (b) Find the total charge contained in a sphere of radius R, centered at the origin. (Do it in two different ways.)

2. Find the electric field inside a sphere that carries a charge density proportional to the distance from the origin, $\rho = kr$, for some constant k. [Hint: This charge density is not uniform, and you must integrate to get the enclosed charge.]

3. A thick spherical shell carries charge density

$$\rho = \frac{k}{r^2}(a \leq r \leq b)$$

, (shown in fig 1). Find the electric field in the three regions: (i) $r < a$, (ii) $a < r < b$, (iii) $r > b$. Plot $|\vec{E}|$ as a function of r, for the case $b = 2a$

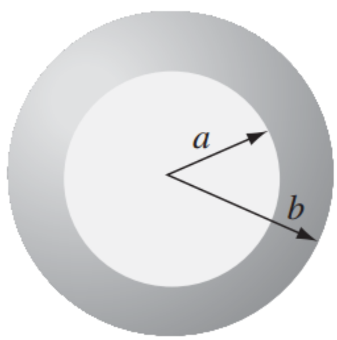

Figure 1

4. Find the potential inside and outside a uniformly charged solid sphere whose radius is R and whose total charge is q. Use infinity as your reference point. Compute the gradient of V in each region, and check that it yields the correct field. Sketch $V(r)$.

5. An infinite plane slab, of thickness $2d$, carries a uniform volume charge density ρ (shown in fig 2). Find the electric field, as a function of y, where $y = 0$ at the center. Plot E versus y, calling E positive when it points in the $+$y direction and negative when it points in the $-$y direction.

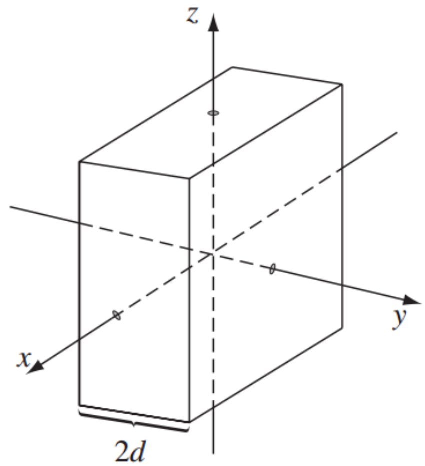

Figure 2